

MEMORIA CIENTÍFICO-TÉCNICA PROYECTOS **INDIVIDUALES**

Convocatoria 2024 - Proyectos de Generación de Conocimiento y actuaciones para la formación de personal investigador predoctoral asociadas a dichos proyectos.

Tipos A, B y RTA

1. DATOS DEL PROYECTO. PROPOSAL DATA

IP 1: Marco Talone

IP 2: Verónica González Gambau

TÍTULO DEL PROYECTO (ACRÓNIMO): Teledetección óptica hiper-espectral para la estimación de la diversidad fitoplanctónica (OPT4DIV)

TITLE OF THE PROJECT (ACRONYM): Hyper-spectral optical radiometry for phytoplankton diversity estimation (OPT4DIV)

2. JUSTIFICACION Y NOVEDAD DE LA PROPUESTA

JUSTIFICATION AND NOVELTY OF THE PROPOSAL

2.1. **Adecuación de la propuesta a las características y finalidad de la modalidad seleccionada. Adequacy of characteristics and the purpose of selected modality.**

OPT4DIV is submitted to the category **Investigación Orientada**.

The research proposed within OPT4DIV is centered on the assessment of marine phytoplankton diversity through hyperspectral remote sensing, and totally aligns with current international efforts and widely consensuated scientific priorities.

At international level, the **United Nations (UN)** has designated the period 2020-2030 as the **Decade of Ocean Science for Sustainable Development**. Among the seven key outcomes of the Decade identified by the UN, **healthy and resilient ocean, productive ocean** and **predicted ocean** are strongly related with the objectives of OPT4DIV. Additionally, OPT4DIV is considered a contribution to the United Nations Sustainable Development Goal (SDG) **14 Conserve and sustainably use the oceans, seas and marine resources for sustainable development**, and specifically a response to the Ocean Emergencies **Coastal Eutrophication** and **Ocean Acidification**.

At European level, numerous specific European Directives are centered on the protection of diversity and biodiversity of the marine environment, among those the **Marine Strategy Framework Directive (MSFD, 2008/56/EC)**, the **Water Framework Directive (WFD, 2000/60/EC)**, and the **Maritime Spatial Planning Directive (MSP, MSFD, 2008/58/EC)**. The relevance of monitoring, protecting and restoring marine ecosystems has been further confirmed by its inclusion among the five greatest challenges to be faced by 2030 and the consequent start of the **EU Mission 3 Restore our Ocean and Waters by 2030**.

Finally, at national level OPT4DIV is framed within the **CSIC's research challenges 6 Producción Primaria Sostenible, 7 Cambio Global**, and especially **13 Océano**.

2.2. **Justificación y contribución esperada del proyecto a la generación de conocimiento en la temática de la propuesta. Hipótesis de partida. Justification and expected contribution of the project to the generation of knowledge on the theme of the proposal. Starting hypothesis**

Many algorithms have been developed to retrieve a variety of satellite products on phytoplankton composition that include phytoplankton functional types (PFTs), size classes, dominant size class, and taxonomically significant phytoplankton pigment concentrations (see [1] and references therein). Nevertheless, the main weaknesses of the current Earth Observation (EO) phytoplankton diversity indicators are their

large uncertainties, especially when phytoplankton is present at low concentrations, and their limitation to discriminate only a few multiple PFTs or groups simultaneously [2], [3], [4]. This is currently limiting our capability to remotely assess and monitor biogeochemical cycles that require a more detailed view of the phytoplankton community such as the carbon cycle, the nitrogen cycle, or the silica cycle, which in turn underlie processes such as the carbon sequestration, or the ocean acidification. This is mainly due to the spectral and radiometric characteristics of past and present multi-spectral satellite payloads, which provide a limited number of spectral bands in the visible wavelength range and medium-to-large bandwidths [5]. In fact, marine hyperspectral radiometry is still a relatively recent Earth Observation field and so far EO sensors with hyperspectral optical payloads such as the Compact High-Resolution Imaging Spectrometer (CHRIS) onboard ESA's PROBA-1, the Hyperspectral Imager for the Coastal Ocean (HICO) and DLR Earth Sensing Imaging Spectrometer (DEGIS) onboard the International Space Station, or the International Space Station, or ASI's Precursore IperSpettrale della Missione Applicativa (PRISMA) have been characterized by low sensitivity in the marine environment [6]. Pioneering satellite missions with hyperspectral payloads specifically designed for marine applications are NASA's Plankton, Aerosol, Cloud, ocean Ecosystem (PACE), launched in 2024, and ESA's Copernicus Hyperspectral Imaging Mission for the Environment (CHIME), planned for 2029. Motivated by the design of these two innovative sensors, great efforts have been made in recent years to develop novel data processing algorithms for estimating phytoplankton diversity in preparation for the data outputs of PACE and CHIME. Nevertheless, concurrent observations of hyperspectral radiometry and phytoplankton community composition are still extremely scarce, and algorithms are mostly based on bio-optical modelling [6].

The objective of OPT4DIV is to bridge the gap between optical sensing and phytoplankton community composition. To do so, it relies on two in situ datasets of hyperspectral radiometry and omics to train a series of algorithms in preparation for the full exploitation of the observations currently provided by PACE as well as future CHIME's.

Main hypotheses underlying OPT4DIV are:

- H1. The higher spectral resolution of satellite hyperspectral sensors will allow a more detailed assessment of phytoplankton diversity than PFTs.
- H2. The rapidly growing dataset of hyperspectral measurements covering different water cases and a number of different ecosystems generated by PACE together with coincident measurements of phytoplankton diversity will allow the development of a robust global algorithm.
- H3. The results of the exploitation of the in situ hyperspectral radiometry and phytoplankton community composition datasets included in OPT4DIV will serve as a convenient basis for the development of a new algorithm directly applicable to the PACE observations.
- H4. The regular and global information provided by satellite observations will enable further investigations to improve our knowledge of key biogeochemical processes such as nitrogen fixation, carbon sequestration, oxic-anoxic remineralization and ocean acidification.

En los proyectos de modalidad de investigación orientada:
In research-oriented projects

2.3. Justificación y contribución esperada del proyecto a solucionar problemas concretos vinculados a la prioridad temática seleccionada. *Justification and expected contribution of the project to solving specific problems linked to the selected thematic priority*

The research proposed in the framework of OPT4DIV is considered a clear contribution to the thematic priority **6 Alimentación, bioeconomía, recursos naturales y medioambiente**. Indeed, the objective of OPT4DIV is to provide

estimates of the phytoplankton diversity that go beyond PFTs based on hyperspectral radiometric satellite measurements.

The availability of this information with the high spatio-temporal resolution provided by satellite remote sensing will be pivotal to achieve a robust assessment of the health of the marine environment. While the bulk marine photosynthetic plankton is a proxy for primary production, a fundamental process that supports higher trophic levels, the specific composition of the phytoplanktonic community is key to unveil a number of biogeochemical processes such as nitrogen and carbon cycles, or ocean acidification, that provide valuable indications of ecosystem-wide dynamics and health. Potential users of high-quality satellite-based estimation of phytoplankton diversity patterns are numerous, ranging from biogeochemical modelers to marine biologists, ecologists and climate scientists.

3. OBJETIVOS, METODOLOGIA Y PLAN DE TRABAJO **OBJETIVES, METHODOLOGY AND WORK PLAN**

3.1. Objetivos generales y específicos. *General and specific objectives.*

The general objective of OPT4DIV is to contribute to the assessment and monitoring of phytoplankton diversity beyond PFTs and to the evaluation of its response to stressors such as climate change or increasing human pressure. Four specific objectives arise from this general one, namely:

- O1. To improve our knowledge of the sensitivity of optical hyperspectral measurements to phytoplankton composition beyond PFTs.
- O2. Develop and validate satellite-radiometry-based indicators to determine and monitor the status and trends of phytoplankton community composition and diversity from hyperspectral observations.
- O3. Identify spatio-temporal patterns of phytoplankton community distribution and diversity in the Mediterranean Sea.
- O4. Assess the main drivers, both natural and anthropogenic, of the spatio-temporal patterns of phytoplankton community distribution and diversity in the Mediterranean Sea.

3.2. Descripción de la metodología. *Description of the methodology.*

From a project management point of view, the work in OPT4DIV is organized into four work packages, the first of which (WP1: Coordination and Outreach) covers the entire duration of the project, while the remaining three (WP2: Algorithm Development, WP3: Algorithm Validation and Improvement, and WP4: Data Exploitation) will be carried out sequentially with large overlaps. Details of each work package are given in the following paragraphs, together with the expected deliverables and the personnel involved (WP leaders are underlined).

WP 1 Coordination and outreach

WP1 will be co-led by IP1 (Marco Talone) and IP2 (Verónica González Gambau). The objective of WP1 is to ensure the proper development of the project and the adherence to the proposed schedule. Many of the participants are physically located at the Instituto de Ciencias del Mar, which greatly facilitates interaction and collaborative work. Regular monthly project meetings are foreseen to complement the internal WP meetings, to keep track of the main progresses and to guarantee a prompt response to any complications. Online participation of the external members of the working group is generally preferred. Communication and outreach activities as well as Data Management will also be part of WP 1. Related details are provided in sections 4.3, 4.4, and 4.5 of this proposal.

Duration: M1-M36

Deliverables: Technical Reports summarizing project management activities and the communication and outreach initiatives: Mid-term report at M18; Final report at M36

Participants: M. Talone, V. González Gambau

WP 2 Algorithm Development

The algorithm development has been organized in two phases:

In the first phase, the recent launch of PACE does not allow to build a sufficiently large and representative database of coincident hyperspectral radiometric observations and phytoplankton community composition measurements. In preparation of such a database, two preliminary algorithms will be developed based on two unique in-situ datasets of coincident hyperspectral radiometric observations and pigment concentration measurements (DB1), and of coincident pigment concentration and phytoplankton community composition measurements (DB2).

The radiometric dataset (DB1) includes more than 150 observations carefully collected in the Mediterranean Sea and the Black Sea in the framework of the BIOMAP project of the European Commission's Joint Research Centre [7] with the collaboration of several researchers, including the IP1. Among a number of key environmental variables, the dataset comprises hyperspectral measurements of sea surface remote sensing reflectance and coincident concentrations of a set of selected pigments. Radiometric measurements were carried out using hyperspectral TriOS (Rastede, Germany) RAMSES radiometers in the nominal range of 320-950 nm with a spectral sampling and resolution of approximately 3.3 nm and 10 nm, respectively, an in-air field-of-view of approximately 7 degrees (radiance sensors), and an integration time varying between 4 ms and 8192 ms as a function of the illumination conditions. Raw data were processed as detailed in [8]. Regarding pigments concentrations, these were obtained through High-Precision Liquid Chromatography (HPLC) analysis of coincident water samples collected approximately at 0.5 m below the sea surface and preconditioned at sea as detailed in [9]. Available pigment concentrations include chlorophyllide-a, chlorophyll-a, chlorophyll-b, chlorophyll-c, carotene-b, alloxanthin, diadinoxanthin, diatoxanthin, 19'-butonayloxyfucoxanthin, fucoxanthin, 19'-hexanoylinoxanthin, peridinin, and zeaxanthin.

The phytoplankton diversity database (DB2) consists of two temporal series from the North-West Mediterranean Sea, one developed along the coast of Blanes (Spain), called the Blanes Bay Microbial Observatory (BBMO, an asset of the ICM-CSIC, <http://bbmo.icm.csic.es/>), and one developed in the bay of Banyuls-sur-mer (France), called Service d'Observation du Laboratoire Arago (SOLA, partly developed by collaborator P. Galand). DB2 consists of a total of 245 near-surface samples (-2 m), 84 were taken bi-weekly in SOLA from January 2015 to April 2016, and 161 were taken monthly in BBMO from 2001 to 2014. In both timeseries, plankton diversity was studied by combining: a) a joint meta-omics approach (here, metabarcoding targeting prokaryotes and eukaryotes, see the specific methods in WP3) with, b) photosynthetic pigment concentrations as measured by HPLC. Metabarcoding includes >10,000 Operational Taxonomic Units (OTUs, equivalent to the species unit) with a taxonomic classification across 7 (for prokaryotes) to 9 ranks (for eukaryotes) ([10], [11]). Based on this taxonomy, a previous effort from collaborator P. Ramond allows to identify phytoplankton taxa and study their patterns at coarse (PFTs) or fine taxonomic resolution (OTUs). The available pigment concentrations are the same as those listed in DB2, representing the bridge between DB1 and DB2, as well as between the two algorithms to be developed. A map of the measurement stations included in the input datasets is shown in Fig. 1.

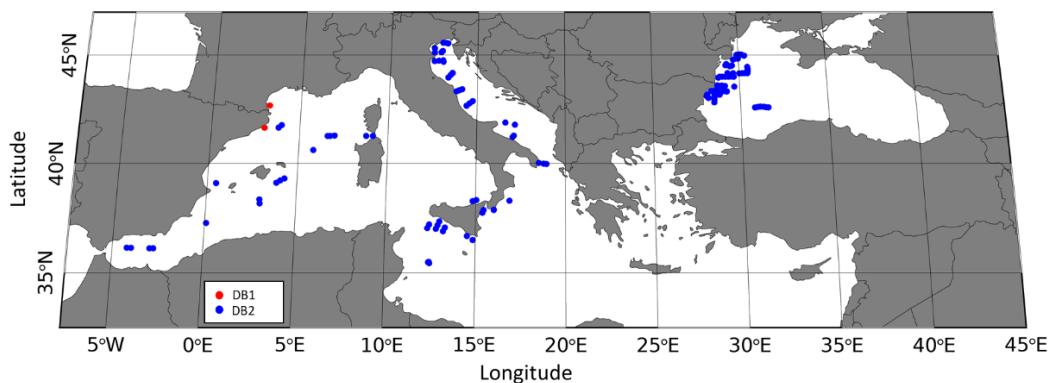


Figure 1. Map of the measurement stations included in the radiometric input dataset (DB1) and the phytoplankton composition dataset (DB2).

Two Artificial Intelligence (AI) models/algorithms will be developed (Task 2.1 and 2.2) and their outputs assembled (Task 2.3). Model training and testing will be performed based on the already available data presented above (DB1 and DB2), while validation will rely on new measurements collected since mid-2024 (see WP3). Part of Task 2.3 is also the fine-tuning of the assembled algorithm based on independent current and future match-ups between PACE observations and in situ water samples. Detail about the Algorithm Validation and the matchup dataset is provided in the next section (WP3).

Task 2.1 Development of the algorithm Radiometry2Pigments

The objective Task 2.1 is to permit the robust retrieval of key pigments from hyperspectral radiometric measurements (called *Radiometry2Pigments*). This will be performed by relying on the aforementioned DB1 dataset and by applying Artificial Intelligence techniques. Both classical Machine-Learning techniques (e.g., Linear Regression, Decision Tree, or Random Forest) and more sophisticated Deep Learning models (e.g., Convolutional Neural Networks or Long Short-Term Memory Networks) will be applied. Foreseen steps are:

- Analysis and regularization of the input dataset;
- Train and test of a series of ML/AI algorithm;
- Evaluation of the model performances with consolidated metrics (e.g. Mean Absolute Error and Mean Percentage Error, Correlation coefficients etc.);
- Assessment of the main drivers of the models among the input parameters.

The expected output of this task is a model allowing to predict photosynthetic pigment concentrations when fed with new hyperspectral radiometric data. The algorithm will be optimized and the quality of its prediction characterized.

Duration: M1-12

Deliverables: Technical Report on the algorithm development and verification of the *Radiometry2Pigments* algorithm at M12. The submission of a scientific paper in a peer-review journal as well as a presentation in a national/international congress documenting this work is foreseen after validation and before M30 (comprehensive publication including all algorithms).

Participants: V. González Gambau, M. Talone, A. Di Cicco, G. Martinez, AI-specialist

Task 2.2 Development of the algorithm Pigments2Diversity

The objective Task 2.2 is to develop a model capable of relating pigment concentrations to phytoplankton diversity beyond PFTs (called *Pigments2Diversity*). Using the taxonomy of the metabarcoding dataset of DB2, the quality of the pigment model prediction will be explored based on different taxonomic resolution from the

PFT to the OTU level (including the Family, Genus, and Species levels). Based on past experience and the literature, the relative abundance of photosynthetic prokaryotic and eukaryotic taxa within the microbial planktonic community will highly correlate with shifts in pigments concentrations [12]. Nevertheless, considering the ecological interactions of phytoplankton with other protists and bacteria, the study will also explore whether the patterns these taxa can be predicted by the pigment model. More specifically, it will focus on 160 abundant OTUs (>1% of the community) considered the main drivers of biogeochemical cycles [13]. It will thus go beyond PFTs by predicting phytoplankton at finer taxonomic resolution (main objective of OPT4DIV), but also by trying to predict the patterns of their abundant non-phototrophic counterparts.

This will be performed by relying on the DB2 dataset and by applying Artificial Intelligence techniques. Both classical Machine-Learning techniques (e.g., Linear Regression, Decision Tree, or Random Forest) and more sophisticated Deep Learning models (e.g., Convolutional Neural Networks or Long Short-Term Memory Networks) will be applied. Foreseen steps are:

- Analysis and regularization of the input dataset;
- Train and test of a series of ML/AI algorithm;
- Evaluation of the model performances with consensuated metrics (e.g. Mean Absolute Error and Mean Percentage Error, Correlation coefficients etc.);
- Assessment of the main drivers of the models among the input parameters.

The expected output of this task is a model allowing to predict phytoplankton diversity patterns at newfound taxonomic resolution when fed with new pigment concentration data. The algorithm will be optimized and the quality of its prediction characterized.

Duration: M1-12

Deliverables: Technical Report on the algorithm development and verification of the *Pigments2Diversity* algorithm at M12. The submission of a scientific paper in a peer-review journal as well as a presentation in a national/international congress documenting this work is foreseen after validation and before M30 (comprehensive publication including all algorithms).

Participants: V. González Gambau, M. Talone, P. Ramond, P. Galand, AI-specialist

Task 2.3 Assembled algorithm and fine-tuning

The objective Task 2.3 is to ensure the seamless assembly of the algorithms developed in Tasks 2.1 and 2.2, thus permitting the direct retrieval of phytoplankton diversity from radiometric measurements. The performance of the assembled algorithm will be evaluated based on an independent dataset of coincident hyperspectral radiometric measurements and omics data currently being collected within the framework of the PETRI-MED project (details are provided in WP3). Foreseen steps are:

- Analysis and regularization of the output dataset of the *Radiometry2Pigments* algorithm with respect to the input dataset of the *Pigments2Diversity* algorithm;
- Evaluation of the assembled model performances with consolidated metrics (e.g. Mean Absolute Error and Mean Percentage Error);
- Assessment of the main drivers of the models among the input parameters.

Duration: M12-M18

Deliverables: Technical Report on the algorithm development, fine-tuning, and verification of the Ensemble algorithm at M18. The submission of a scientific paper in a peer-review journal as well as a presentation in a national/international congress documenting

this work is foreseen after validation and before M30 (comprehensive publication including all algorithms).

Participants: V. González Gambau, M. Talone, A. Di Cicco, G. Martínez, P. Ramond, P. Galand, AI-specialist

WP 3 Algorithm Validation

The algorithm validation has two main components, namely the data collection (Task 3.1) and the data analysis (Task 3.2).

Task 3.1 Data collection

A novel dataset will be created to validate the predictions of the assembled algorithms developed in WP2. This will encompass radiometric measurements provided by PACE and further omics measurements collected at the SOLA and the BBMO timeseries from mid-2024 on. Additionally, measurements performed in different locations of the Mediterranean Sea and the Atlantic Ocean will be incorporated, collected thanks to the participation in oceanographic cruises (for which the ICM-CSIC offers further opportunities).

Regarding SOLA and BBMO, regular water sampling activities are foreseen weekly to monthly within the framework of other independent projects. Samples are generally processed within 12 months from the water sampling. Concerning the oceanographic cruises, some have already been performed in the framework of the PETRI-MED project (see section 3.5 *Previous results of the team in the theme of the proposal*), namely:

- 10-day cruise onboard the Spanish Vessel B/O García del Cid including 31 measurement stations of concurrent omics, pigment concentrations, and in situ hyperspectral (TriOS RAMSES) measurements in September 2023 (PI M. Talone).
- 10-day cruise onboard the Spanish Vessel B/O Angeles Alvariño including 10 measurement stations of concurrent omics, pigment concentrations, and in situ hyperspectral (TriOS RAMSES) measurements in October 2024 (PI E. Fraile).

Others are planned, specifically:

- 28-day cruise onboard the Italian vessel Gaia Blu in the framework of the Italian ITINERIS project in July 2025 (PI Emanuele Organelli, invitation letter: <https://saco.csic.es/s/9HsP47jG5fq7Aet>).
- 20-day cruise in the framework of the GOOSE in September 2026 (PIs Rafel Simó and Martí Galí, invitation letter: <https://saco.csic.es/s/554yMKd9nE3XEBr>).
- Various cruises onboard the Spanish vessel B/O Angeles Alvariño in the framework of the Ocean Color Satellite Vicarious Calibration project in 2026-2027 (PI Eugenio Fraile, invitation letter: <https://saco.csic.es/s/R78WsJTGeAkB2wt>).

Data collection will follow consolidated protocols [14]. Regarding hyperspectral radiometry, this will be carried out using hyperspectral TriOS (Rastede, Germany) RAMSES radiometers [15], namely one ACC irradiance sensor operated on a 6 m telescopic mast installed on the upper deck of the ship, and one or two ARC radiance sensors operated at a nominal depth of approximately 2 cm on a towed T-shaped floating frame named Optical Floating System (OFS). As displayed in Figure 2, OFS has the capability to simultaneously produce radiometric measurements in combination with ancillary tilt/compass data. During measurement stations, simultaneously with OFS radiometric data, spectral beam attenuation, absorption coefficients, and backscattering coefficient will be measured with a SeaBird AC-S meter (Bellevue, WA) and a Sequoia Hyper-bb meter (Bellevue, WA).



Figure 2. Optical Floating System (OFS) used to collect hyperspectral radiometric measurements. The RAMSES-ARC radiometers are located in the middle of the main arm. The tilt/compass unit is located on its rightmost end.

Additionally, surface water samples will also be collected to determine pigments concentration through High Performance Liquid Chromatography (HPLC). As for omics measurements, the protocols deployed along the BBMO and SOLA timeseries will be followed. Briefly, surface seawater will be sampled with a niskin bottle, 5-10 L will be filtered using a peristaltic pump sequentially through a 20 μm and 3 μm pore-size polycarbonate filters, followed by a 0.2 μm pore-size GV Sterivex cartridge (Merck-Millipore, Darmstadt, Germany). The filters will be frozen and stored at -80°C until further processing. Back in the laboratory, the planktonic DNA will be extracted in a lysis buffer and following the protocol of the Qiagen AllPrep kit (Qiagen, Hilden, Germany). Specific taxonomic gene markers will be targeted and PCR-amplified using the universal primer set developed in [16] and the eukaryotic specific primer set of [17]. Sequencing will be performed with an Illumina MiSeq sequencer at the Centro Nacional de Análisis Genómico (CNAG-CRG, Barcelona, Spain). Finally, the metabarcoding datasets will be built following the DADA2 pipeline [18] and using the ICM-CSIC's own Marine Bioinformatics Platform (MARBITS). Taxonomic annotation will be performed using the SILVA v138.1 and PR2 v5.0.0 databases, and phytoplankton taxa identified. Following, the same protocols and pipelines will allow the seamless aggregation of this new data to the existing collection (DB2).

Duration: M1-M30
Deliverables: Technical Report summarizing all collected field measurements relevant to OPT4DIV at M30. Dataset collected within the framework of OPT4DIV will be pushed in an open-access repository as detailed in section 4.6 *Summary's management plan of the planned data*.
Participants: M. Talone, G. Martínez, M. Lloret, J. Salvador, G. Volpe, G. Zibordi

Task 3.2 Data analysis

Data collected as part of Task 3.1 will be used to validate all the algorithms developed in WP 2. Specifically, model performances will be evaluated based on consolidated metrics (e.g. Mean Absolute Error and Mean Percentage Error, Correlation coefficients etc.) and compared to other published algorithms, e.g., by grouping back to the PFT level, or in terms of diversity index (currently being developed within PETRI-MED).

Duration: M18-M30
Deliverables: Technical Report on the results of the validation activities. The submission of a scientific paper in a peer-review journal as well as a presentation in a national/international congress documenting this work is foreseen at M30 (comprehensive publication including all algorithms).

Participants: M. Talone, V. González Gambau, A. Di Cicco, G. Martínez, P. Ramond, P. Galand, AI-specialist

WP 4 Data Exploitation

Following the algorithm validation, a preliminary exploitation of phytoplankton diversity maps is foreseen. Based on data provided by PACE, regional and basin-level seascape spatial assessments will be conducted by combining phytoplankton diversity estimates with in situ measurements as well as physical and biogeochemical indicators of key oceanographic processes.

Task 4.1 Spatio-temporal analysis

In this task, regional and basin-level seascape spatial variability will be assessed. This will be done by applying techniques like Principal Component Analysis, Cluster Analysis, and Semivariogram Analysis ([19]) with the main objective of evaluating the potential of remote sensing in supporting the identification of fragmented habitats and the recognition of new zones or ecological corridors. To highlight spatial correlation, singularity analysis will be conducted. Singularity exponents are dimensionless variables providing information about the local regularity of a certain signal at any given point. When applied to remote sensing variables, singularity exponents are strongly representative of the horizontal flow, providing a very useful tool to assess the geophysical consistency ([20], [21]). Due to the spatial distribution of DB1 and DB2, the analysis will be restricted to the Mediterranean Sea.

Duration: M24-M36

Deliverables: Technical Report on the spatio-temporal analysis at M36. The submission of a scientific paper in a peer-review journal as well as a presentation in a national/international congress documenting this work is foreseen at M36 (related with all data exploitation activities).

Participants: V. González Gambau, M. Talone, G. Martínez, A. Di Cicco

Task 4.2 Drivers evaluation

From the results obtained in Task 4.1, a detailed analysis will be performed to identify the main drivers of remotely sensed phytoplankton diversity. This will include both natural processes such as extreme events like floods or marine heat waves and anthropogenic processes such as increased human pressure in urban or touristic areas. The different remotely sensed variables will be preprocessed to highlight main spatio-temporal patterns ([19], [20], [21]) and then related to phytoplankton diversity. Analogously to task 4.1, due to the spatial distribution of DB1 and DB2, the analysis will be restricted to the Mediterranean Sea.

Duration: M24-M36

Deliverables: Technical Report on the drivers analysis at M36. The submission of a scientific paper in a peer-review journal as well as a presentation in a national/international congress documenting this work is foreseen at M36 (related with all data exploitation activities).

Participants: M. Talone, V. González Gambau, G. Martínez, P. Ramond, P. Galand

3.3. Plan de trabajo y cronograma. Work plan and schedule.

Work Breakdown Structure and Project Schedule are detailed in figures 3 and 4, respectively.

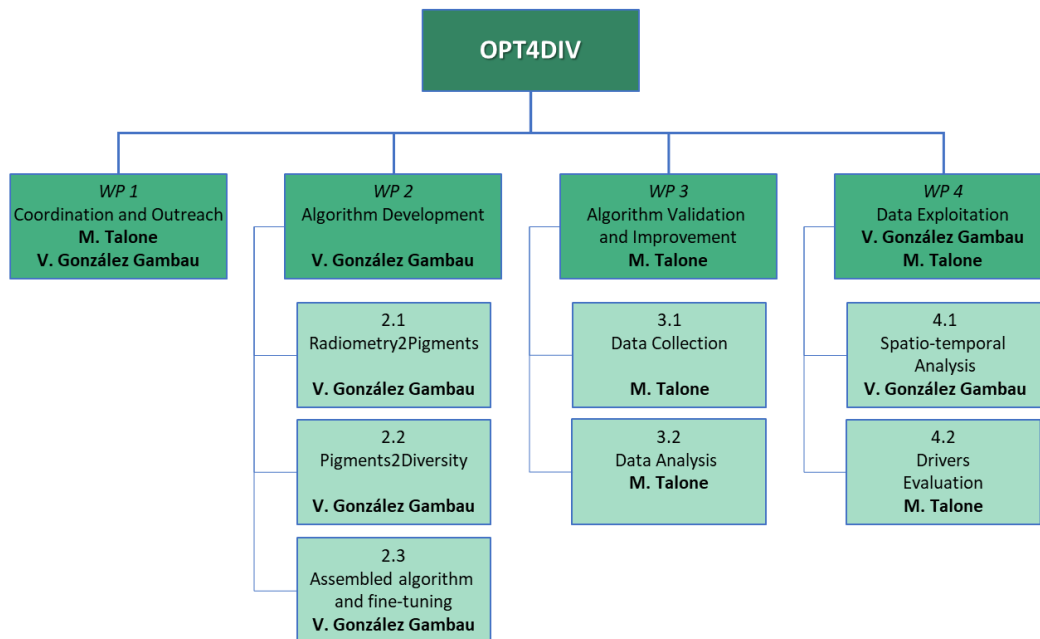


Figure 3. Work Breakdown Structure of OPT4DIV

		Y1				Y2				Y3			
Work Packages	Leader	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
WP 1 Coordination and Outreach	MT + VGG												
WP 2 Algorithm Development	VGG												
2.1 Algorithm Development – Radiometry2Pigments	VGG												
2.2 Algorithm Development – Pigments2Diversity	VGG												
2.2 Assembled Algorithm and Fine-Tuning	VGG												
WP 3 Algorithm Validation	MT												
3.1 Data Collection	MT												
3.2 Data analysis	MT												
WP 4 Data Exploitation	VGG + MT												
4.1 Spatio-Temporal Analysis	VGG												
4.2 Drivers Evaluation	MT												

Figure 4. Project scheduling of OPT4DIV

3.4. Identificación de puntos críticos y plan de contingencia. *Identification of critical points and contingency plan.*

The main critical point for the successful execution of OPT4DIV is its strong dependence on new measurements for the validation and fine-tuning of the developed algorithms. For the in situ component, OPT4DIV relies on in situ measurements from both marine buoys (phytoplankton community composition and pigments (SOLA only)) and measurement campaigns (radiometry and phytoplankton community composition). While data availability from marine buoys is assured, both the ship availability and the weather conditions during the campaigns may be an issue. Ship availability is beyond our control, but most of the planned activities have already been approved and it is reasonable to assume that they will be carried out. Regarding the weather conditions during the campaigns, the planned activities already have a rather wide temporal interval (in the order of 20 days), which should ensure a minimum number of measurements even in case of unfavorable weather conditions. Regarding the satellite component, PACE has been providing observations since April 2024 and no technical problems have been reported so far. In case of satellite failure, in situ measurements could be used, although with a consequent reduced statistical representativeness of the validation dataset. In this

very unfavorable case, the exploitation of the measurements provided by PRISMA can be envisaged, still WP4 will be strongly reduced.

Main potential risks affecting the proper development of OPT4DIV are summarized in Table 1, jointly with an estimate of their severity and likelihood, and a list of proposed solutions.

Table 1. Contingency Plan of OPT4DIV

Risk [Severity/Likelihood]	Proposed Solution
Interdependency between Work Packages [Medium/Medium]	Work Packages 2, 3, and 4 are performed sequentially. Workload will be remodulated and redistributed among research group and working group members to cope with possible delays or difficulties.
Difficulties of recruitment [Medium/Low]	Massive dissemination of the vacancy counting on the joint networks of the members of PETRI-MED will be done, a remodulation of the workload among research group and working group members is foreseen to cope with possible delays or difficulties.
Mobility reduction due to unpredictable events [High/Unknown]	The cancellation of all field activities would seriously affect the algorithm validation and improvement. The rest of the project activities, instead, do not require in-person meetings and can be performed online.
Oceanographic campaign constraints [Medium/Medium]	Planned campaigns are considered long enough to ensure a minimum number of stations even in case of unfavorable weather or technical issues. Their complete cancellation instead would partially affect the algorithm validation and improvement, which would then rely only on the measurements provided by SOLA and BBMO with a consequent reduced statistical and geographical representativity.
Satellite failure [Very high/Low]	The availability of satellite hyperspectral measurements is key for the algorithm validation and improvement. In case of satellite failure, in situ measurements can be used considering the consequent reduced statistical representativity.
In situ sensors failure [Low/Low]	High redundancy of the in situ sensors is guaranteed by the collaboration with scientists from the Italian Consiglio Nazionale delle Ricerche and the European Commission's Joint Research Centre who own similar and totally interchangeable instrumentation.
Insufficient data for the fine-tuning of the AI algorithm [High/Medium]	The future validation dataset based on PACE matchups is key for OPT4DIV. In case the amount of collected data is considered insufficient, data from freely available on open repositories such as EMODnet, Zenodo or PANGAEA will be included.

3.5. Resultados previos del equipo en la temática de la propuesta. *Previous results of the team in the theme of the proposal.*

OPT4DIV is the evolution of a long series of efforts of the proposing team towards the algorithm development, data processing and analysis, validation, and exploitation of satellite measurements.

The IP1, **Marco Talone**, has worked for more than 10 years on the characterization of hyperspectral radiometers and the consolidation of in situ measurement protocols for satellite product validations ([8], [15], [22], [23], [24], [25], [26], [27], [28]). During his time at the European Commission's Joint Research Centre, he contributed to the creation of the input dataset DB1 by collecting and processing all included radiometric measurements. He is deeply involved in the supporting activities to Copernicus Sentinel-3 being a member of the Ocean Color Sentinel 3 Validation Team and co-leading the two measurement stations of the Ocean Color component of the Aerosol Robotic Network (AERONET-OC) in Spanish waters, namely the Casablanca and PLOCAN sites ([29], [30], [31], [32], [33], [34]). AERONET-OC is currently the main provider of high-quality radiometric measurements for satellite product validation and produces more than 90% of the measurements used by ESA and NASA for that purpose. M. Talone also coordinated two campaigns of the BIOMAP program, performed in the western Mediterranean Sea in 2021 and 2023. Particularly relevant for the development of OPT4DIV is the project PETRI-MED (*Phytoplankton biodiversity through remote sensing and omics in the Mediterranean Sea, 04/23-04/27*), financed through the Biodiversa+ 2021-2022 BiodivProtect call coordinated by M. Talone, and executed by an international consortium including the Spanish ICM-CSIC and *Institut de Recerca i Tecnologia Agroalimentàries*, the French *Centre National de la Recherche Scientifique*, the Italian *Consiglio Nazionale delle Ricerche*, the Slovenian *Nacionalni Institut za Biologijo*, and the Israeli *University of Haifa*. Though sharing similar objectives, in fact, OPT4DIV and PETRI-MED apply complementary methodologies. The main difference between both projects is that PETRI-MED relies on multi-spectral satellite

radiometry, as opposite to OPT4DIV that is based on the exploitation of hyperspectral data. The limited spectral resolution of the multi-spectral observations does not permit to build univocal relations between ocean color measurements and pigment concentrations, consequently plankton diversity in PETRI-MED is based on omics data instead of pigment analysis. To compensate for the lack of spectral information, a more ecological approach is adopted by PETRI-MED to retrieve plankton diversity by combining satellite radiometric measurements with modelled auxiliary parameters such as nutrients concentrations, sea surface temperature, salinity, and mixed-layer depth, with the aid of Artificial Intelligence modelling. A similar approach will be adopted in OPT4DIV in Work Packages 2.1 and 2.2, building on the expertise acquired during PETRI-MED. M. Talone will co-coordinate WP 1 (Coordination and outreach) and WP4 (Data exploitation) and will lead WP 3 (Algorithm Validation).

The IP2 **Verónica González-Gambau** has worked for more than 15 years on the characterization and calibration of satellite radiometers and algorithm developments for in-orbit corrections of radiometric measurements ([35], [36], [37], [38], [39], [40]). She has also a long experience in the development of algorithms to retrieve geophysical parameters over the ocean from radiometric observations, such as the global and regional Sea Surface Salinity maps in the Baltic Sea ([41], [42]), Black Sea ([43], [44], [45]), Southern Ocean ([46]) and Arctic Ocean ([37]), distributed by the Barcelona Expert Center (BEC). She is the responsible at ICM of the ESA Expert Support Laboratories of Level 1 of the SMOS mission. She is currently involved in the Level 2 algorithm development of oceanic variables in the upcoming ESA's CIMR (Copernicus Imaging Microwave Radiometer) mission, planned for 2029. V. González Gambau will co-coordinate WP1 (Coordination and outreach) and WP4 (Data exploitation) and leads WP 2 (Algorithm Development). The member of the research group **María Isabel Lloret Fernández** has a more than 25 years of experience in supporting marine research activities at ICM having participated in a number of national and international projects. Her contribution to the project will be centered on the preparation and execution of the foreseen measurements during the oceanographic campaigns (WP 3.1).

The member of the research group **Joaquín Salvador Castiella** has a more than 20 years of experience in supporting marine research activities at ICM having participated in a large number of national and international projects. Similar to María Isabel Lloret Fernández, his contribution to the project will be centered on the preparation and execution of the foreseen oceanographic campaigns (WP 3.1).

The member of the working group **Gonzalo Martínez Fornos** is an environmental scientist with a master in remote sensing. He is currently pursuing his PhD focusing on the analysis of the biota of the Mediterranean Sea under the supervision of the PI Marco Talone and Elisa Berdalet Andrés. G. Martínez will support WP2 (Algorithm Development) WP3 (Algorithm Validation), and WP4 (Data Exploitation).

The external members of the working group cover a wide range of competencies of extreme relevance for the project: **Giuseppe Zibordi**, formerly at the European Commission's Joint Research Center, is currently with the NASA Goddard Space Flight Center in charge of the scientific validation and evolution of AERONET-OC; he has 30 years of experience in experimental optics in support of satellite applications. **Gianluca Volpe** and **Annalisa Di Cicco** have a consolidated expertise in operational oceanography and exploitation of satellite radiometry. Both are members of the Copernicus Marine Service and authors of the algorithms implemented in the currently provided products of Chlorophyll-a concentration [47] and Phytoplankton Functional Types abundances [48], respectively. **Pierre Galand** and **Pierre Ramond** are research director and post-doc fellow at LECOB-CNRS, respectively. Their expertise is focused on microbial ecology, especially high-throughput sequencing applied to study planktonic communities. Annalisa Di Cicco, Pierre Galand, and Pierre Ramond are members of PETRI-MED.

Previous projects involving members of OPT4DIV and related with its research subject include:

PETRI-MED, BIODIVERSA+, 2023-2027; *PTI TELEDETECT*, CSIC, since 2020; *INTERACT*, Spanish National Research Program, 2021-2024; *OPT4CYAN*, Spanish National Parks Autonomous Agency, 2021-2024; *AERONET-OC*, European Commission's Joint Research Centre, since 2019; *BIOMAP*, European Commission's Joint Research Centre, 2021 and 2023; *Assessment of the JRC Free-Fall profiler hyperspectral data applying the CSIC optical processor*, European Commission's Joint Research Centre, 2024; *Assessment of Near Surface Radiometry measurements*, European Commission's Joint Research Centre, 2023; *Sentinel-3's OLCI Bidirectional Reflectance Distribution Function correction*, European Organization for the Exploitation of Meteorological Satellites, 2022-2024; *SMOS Expert Support Laboratory*, European Space Agency, since 2003; *L-BAND*, Spanish National Research Program, 2018-2020; *PROMISES*, Spanish National Research Program, 2016-2017; *Southern Ocean Freshwater*, European Space Agency, 2021-2023; *Baltic+ Salinity Dynamics*, European Space Agency, 2018-2020; *EO4SIBS*, European Space Agency, 2019-2021; *Arctic+ Salinity*, European Space Agency, 2018-2020; *Arctic+ Salinity CCN*, European Space Agency, 2023-2024; *CIMR Level-2 Product and Algorithm Development*, European Space Agency, 2023-2027; *CMEMS-OCTAC*, Copernicus, EC, since 2015 (CNR); *CMEMS-MOBTAC*, Copernicus, EC, since 2015 (CNR).

3.6. Recursos humanos, materiales y de equipamiento disponibles para la ejecución del proyecto. *Human, material and equipment resources available for the execution of the Project.*

Most of the members of OPT4DIV are with the Barcelona Expert Centre (BEC, <https://bec.icm.csic.es>), a joint initiative of CSIC, Universitat Politècnica de Catalunya and Institut d'Estudis Espacials de Catalunya created in 2007 to provide support to the European Space Agency Soil Moisture and Ocean Salinity (SMOS) data processing and algorithm development. Ever since, BEC has grown into a recognized research and development laboratory in remote sensing (from level 0 to level 4) and a data production and distribution center for remote-sensing products through the Scientific-Technical Service BEC-RS. Besides internal staff, to ensure the proper development of the project and the fulfillment of its objectives, a specialist in artificial intelligence and data processing is requested, namely **a 2.5-year Titulado Superior FC1 contract** (55.8 K€/year + 8 K€ end-of-contract indemnity), whose support is required in Tasks 2.1, 2.2, 2.3, and 3.2.

Regarding computational resources, OPT4DIV will rely on the Scientific-Technical Service "BEC-RS" and "Bioinformatics". A first estimate of the needs of OPT4DIV based on 2024's catalogue includes:

- *Software Development for remote sensing data processing/exploitation*
Task: Development and test of a program to generate a list of match-ups with PACE
Estimated effort/cost: 60 units, equivalent to 60 person*hour / 0 euros.
- *Remote sensing data processing*
Task: Download, extraction and pre-processing of satellite data
Estimated effort/cost: 32 units, equivalent to 2000 hours on a dedicated 16-core workstation / 181,12 euros.
- *Data storage and visualization*
Task: Safe storage of satellite data during the execution of the project
Estimated effort/cost: 72 units, equivalent to 2 TB *36 months / 175.68 euros.
- *Bioinformatic data processing*
Task: Process sequenced DNAs

Estimated effort/cost: 100 units = 100 KCPU*h / 566 euros.

As for in situ sensors, for the optimal development of OPT4DIV, access to all the equipment of the Ocean Color Group of the European Commission's Joint Research Centre has been granted. This includes multi-spectral and hyperspectral radiometers (Satlantic microPRO, Biospherical C-OPS, and TriOS RAMSES). Absorption-attenuation and backscattering sensors (Seabird AC-S and Sequoia Hyper-bb), sun-photometer (MicroTOPS) and filtration set (Sartorius) are available at the ICM-CSIC.

Finally, it is mentioned that the ICM-CSIC, as a Severo Ochoa Center of Excellence, counts on dedicated Units to support researchers throughout the life cycle of their projects to ensure research viability and scientific excellence. Namely, the Project Management Unit and the administrative department offers a team of professionals that facilitates the scientific and economic follow-up of the projects; the Talent Development and Work Environment Units contribute with high-level training programs and a mentoring program to develop staff talent at all seniority levels, and the Outreach & Communication Unit assists in the design and implementation of the project dissemination and communication plan.

4. IMPACTO ESPERADO DE LOS RESULTADOS. *EXPECTED IMPACT OF THE RESULTS.*

4.1. **Impacto esperado en la generación de conocimiento científico-técnico en el ámbito temático de la propuesta. *Expected impact on the generation of scientific-technical knowledge in the thematic area of the proposal.***

Expected impact of the knowledge generated within OPT4DIV in the thematic priority **6 Alimentación, bioeconomía, recursos naturales y medioambiente** is manifold. The results of OPT4DIV will contribute to demonstrate the added value of satellite mission embarking hyperspectral optical payloads specifically designed for marine applications such as NASA's PACE and the future ESA's CHIME, as well as to improve our understanding of critical biogeochemical processes such as carbon or the nitrogen cycles, or ocean acidification [5].

Phytoplankton plays an important role in the **carbon cycle**. However, diverse phytoplankton groups affect the carbon cycle differently. Additionally, the time lag between carbon sequestration and its reemission into the atmosphere strongly depends on the phytoplankton size. A better estimation of composition and distribution of the phytoplankton community would largely improve carbon budget estimations and forecasts. Phytoplankton also participate in the **nitrogen cycle**: phytoplankton incorporate nitrogen mainly as ammonia and nitrate, but only a specific group, the cyanobacteria, interacts with nitrogen in its gaseous form. Nitrogen is key to support photosynthesis and primary production, so the possibility of robustly distinguishing between cyanobacteria and the other classes would benefit ecological modelling and fish stock predictions.

Additionally, distinct phytoplankton species affect and are affected differently by the sea water alkalinity, notably coccolithophores can produce a decrease of the sea water alkalinity when forming calcium carbonate plates around their cell, while the same process is hindered by **ocean acidification**, with strong repercussions on the carbon cycle. Again, the possibility of identifying phytoplankton species can drastically improve our understanding of the ecosystem health triggering a number of ecological and climatological studies.

Finally, exploring our ability to predict patterns of non-photosynthetic microbes and protists from space is a long-shot venture. Still, covering even some of the other members of the microbiomes may yield new and manifold insights into the biogeochemical cycles they are driving at the global oceanic scale. Such knowledge is crucial to improve our understanding of the functioning of the global ocean and predict the impact of climate change on its ecosystems. This has been undertaken

before but still represents an important knowledge gap [49], ensuring that new results in this direction could have a high-impact for the field.

4.2. **Impacto social y económico de los resultados previstos. *Social and economic impact of the expected results.***

Protection, monitoring, and restoration of marine diversity and biodiversity are key matters in international directives and recommendations as the **EU Biodiversity Strategy for 2030** (2020) or the **European Green Deal Call** (2020) [Topic: **Restoring biodiversity and ecosystem services**] the **Convention on Biological Diversity** (CBD-2010, 2020) including the **Aichi Biodiversity Target**, the **Sustainable Development Goals of the UN Agenda 2030** (Resolution 70/1), the **UN Decade of Ocean Science for Sustainable Development 2021-2030** (Resolution 72/73), and the establishment and long-term support of the **Intergovernmental Platform on Biodiversity and Ecosystem Services (IPBES)**. Specifically, the EU Biodiversity Strategy for 2030 emphasizes that the current network of Marine Protected Areas (MPAs) is insufficient to guarantee the restoration, resilience and protection of the European ecosystems, recommending an increase of the monitored and protected areas with specific focus on the **identification of ecological corridors among Marine Protected Areas**, stressing the need of a **quantitative evaluation of the progress towards recovery**. On the same line, while the MSFD includes Biodiversity as Descriptor 1 of GES, the report on its implementation (COM/2020/259 final) emphasizes the need for a coherent and quantitative determination of GES leading to an **objective assessment of the effects on the marine environment of the measures put in place by member states**, and call for innovative cost-effective monitoring systems that would allow broad spatial and temporal coverage of the regional seas. Finally, the assessment of ecological status is supported under the WFD by the so-called Intercalibration Decision. Within this context and with the objective of supporting biodiversity conservation and restoration policies at both European and national levels, research developed within OPT4DIV will contribute to provide policy makers and other stakeholders with the adequate knowledge to:

- Permit the adoption of **prioritization approaches to ecosystem management** based on quantitative and real-time metrics.
- **Design and implement different protection strategies and policies** to protect ecosystem biodiversity.
- **Quantify the results** of implemented actions at both European, basin and local levels.
- Enable **systematic, fact-supported, and area-based management** of MPAs, Key Biodiversity Areas (KBAs), and Ecologically or Biologically Significant Marine Areas (EBMSAs).
- Determine existing (or lacking) **interactions and complementarity between MPAs** through different mechanisms (e.g. UNESCO Biosphere Reserve, Ramsar sites, private protected areas, etc.) thanks to a synoptic view of the plankton biodiversity.

4.3. **Plan de comunicación científica e internacionalización de los resultados (indicar la previsión de publicaciones en acceso abierto). *Plan for scientific communication and internationalization of the results (indicate the forecast of open access publications).***

Scientific results arising from OPT4DIV will be published in **open-access** whenever possible, also benefitting from total or partial financing of the Article Processing Charges offered by CSIC. At least two, likely three, publications in high impact peer-reviewed journals are expected. Participation in relevant **international congresses and workshops** is also foreseen to diffuse the results of the project. Complementary to scientific publication and congresses, the **inclusion in the EU Mission Restore our Ocean and Waters Charter** and the **endorsement by the United Nations**

Decade of the Ocean Science for Sustainable Development will be requested with the objective of gaining access to their specific international workshops and match-making events, which resulted particularly useful in previous projects. The strong collaboration with the European Commission's Joint Research Center (**JRC**), the US National Aeronautics and Space Administration (**NASA**), the Italian Consiglio Nazionale delle Ricerche (**CNR**) and the French Centre National de la Recherche Scientifique (**CNRS**) as well as the tight relation with the European Union's **Copernicus Marine Service** currently coordinated by CNR will definitely increase the visibility of OPT4DIV and the internationalization of its results. The results of OPT4DIV will be presented to public agencies and stakeholders at dedicated meetings, particularly with ESA and EUMETSAT, to better position the proposing team to compete for contracts such as Invitations to Tender (ITT) or Horizon Europe projects. Finally, it is mentioned that both the Copernicus Marine Service and NASA are directly involved in the working group of OPT4DIV, while the European Commission's Joint Research Centre (<https://saco.csic.es/s/R78WsJTGeAkB2wt>), the **European Space Agency** (<https://saco.csic.es/s/R5fBQAKL9WiQfE>), and the **PACE Science Team** (<https://saco.csic.es/s/MjGeMtE2PZWDgsN>) have formally expressed their endorsement to the project.

4.4. Plan de divulgación de los resultados a los colectivos más relevantes para la temática del proyecto y a la sociedad en general. *Plan for dissemination of the results to the most relevant groups for the theme of the project and to society in general.*

Expected knowledge generated by OPT4DIV will be shared with stakeholders through **technical reports** (e.g., targeted guidelines and policy briefs), and periodic mailing. Members of OPT4DIV participate in several research networks, such as CSIC's PTIs Teledetect and PolarCSIC, or the Spanish network of optical proximal sensing (NetOPS). Additional possible stakeholders are the end users of the Barcelona Expert Centre, which webpage has an average of 2.000 unique visitors and 30 GB of downloaded data monthly.

As for the dissemination to the general public OPT4DIV will make the most of the **institutional dissemination platforms** of all the contributing institutions coordinating simultaneous press releases as well as their link/re-tweet among research institutes and participating networks. In addition to social media, **institutional periodic bulletins** and ad-hoc websites (e.g., CSIC's *ICM Divulga* and CNR's *Almanacco della Scienza*) will be employed. Members of OPT4DIV participate in a number of **outreach activities**, such as *Pint of Science*, aimed to the general public of all ages, the *microOcean* project for high-school students in collaboration with the NGO *EcoOcean*, in the *Oceánicas* meetings, the *Oceans and Human Health Chair*, or *EU4Ocean*. Activities are foreseen to be organized during the European Researcher's Night event, generating awareness about microscopic life in the oceans and why it is essential. With the support of the Communication and Outreach Unit, **high and elementary schools will be contacted**, and talks will be planned in advance. Coastal neighborhoods will be also engaged with general talks on the importance of microbial marine life in a healthy ocean and how this aligns with European Green-Deal strategies.

En los casos en que sea de aplicación: In cases where it is applicable

4.5. Plan de transferencia y valorización de los resultados. *Transfer plan and valorization of results*

Expected knowledge transfer from OPT4DIV is mostly related to developed algorithms and to their potential use to generate operational satellite products, which can be certainly managed by the Barcelona Expert Centre. Additionally, this will help positioning the project team to capitalize on future opportunities within the upcoming CHIME mission, including involvement in the Mission Advisory Group and

participation in the mission's preparatory activities (such as campaigns and development of retrieval algorithms).

It is mentioned that OPT4DIV can count on the assistance from the ICM Technology Transfer Unit to ensure the maximum valorization of research outputs.

4.6. **Resumen del plan de gestión de datos previsto. *Summary's management plan of the planned data***

Access right to the input dataset is subject to the decision of a number of researchers that contributed to its creation and exceeds PIs authority. However, PIs commit that all data produced within the framework of OPT4DIV will follow the **FAIR** principles. They will be **findable** by submitting them to general-purpose open-access repositories like EMODnet, Zenodo, and/or PANGAEA to get a permanent identifier. Accompanying metadata will be identified by keywords to optimize indexing and possibility for harvesting by stakeholders. They will be **accessible** through the same open-access repositories. Data will be **interoperable** by following the standards of the field to allow data exchange and re-use. Accompanying readme files on methods and codes will ensure that the data are **reusable**. Data storage during the project development will be implemented relying on local resources. **Redundancy policies** (RAID5), **monthly backups** and cloud storage are foreseen to guarantee data safety and prevent its loss. At the end of the project, data will be published with a **Digital Object Identifier under Open Access** through DIGITAL.CSIC. Long term security will be ensured by their inclusion in the aforementioned repositories free of charge. The data manager of OPT4DIV is the IP1 (M. Talone).

4.7. **Efectos de la inclusión de género en el contenido de la propuesta. *Effects of gender inclusion in the content of the proposal***

PIs, research team and working group commit themselves **to offer gender, intercultural, social, and age-sensitive working conditions** and to implement the necessary measures to **guarantee equal opportunities of integration into all levels of the project** between people of identity diversity (gender, cultural, social and age). Efforts will be made to guarantee that this equality is effective at all the project phases (formulation, implementation, evaluation and dissemination), for example, in terms of **access to resources, authorship, and participation in congresses**, aiming at promoting positive role models in science and at the same time, helping to deconstruct stereotyped prejudices. It is mentioned that the Institute of Marine Sciences abides by the Equality Plan of the CSIC and the Equality Plan of the ICM, promoted by the **Equality and Diversity working group of the ICM** (<https://gep.icm.csic.es/>). Finally, it is noted that the women/men proportion is 50/50 at both PI and research group levels while it is 30/70 at working group level.

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5. JUSTIFICACIÓN DEL PRESUPUESTO SOLICITADO. *JUSTIFICATION OF THE REQUESTED BUDGET*

To ensure the fulfillment of the objectives of OPT4DIV and allow its proper development a **total budget of 202,500€ is requested**, subdivided as follows.

Staff: A specialist in Artificial Intelligence modelling is requested, namely a 2.5-year *Titulado Superior FC1* contract (55,800 €/year + 8000 € end-of-contract indemnity), whose support is required in WPs 2.1, 2.2, 2.3, and 3.2: **foreseen budget 147,500 €**

Consumables: Consumables will be necessary for the participation in oceanographic campaigns, including filters, dry ice, and chemicals. **foreseen budget 12,500 €**

Minor equipment: The purchase of a laptop to be assigned to the hired AI specialist is planned, plus external disks for data storage and exchange: **foreseen budget 2,000€**

Publications: At least two publications in high impact journals are expected from OPT4DIV, publication costs will be partially covered by CSIC: **foreseen budget 3,000 €**

Travels: Participation in international congresses will be encouraged, especially to younger scientists, to diffuse the results of the project (3,000 €/year). Additionally, staff travels related to the participation to the oceanographic campaigns is foreseen (5,000 €): **foreseen budget 14,000 €**

Others: Budget will be needed to support the analysis of campaign measurements (extraction and sequencing). Considering 250 stations in total for an approximate price of 50 € per sample, the associated budget is 12,500 €. Satellite and omics data processing through the CSIC's SCT has been estimated at approximately 1,000 €. Finally, shipping of the instrumentation used and the samples collected in the measurement campaigns must be considered (10,000 €): **foreseen budget is 23,500 €**

6. CAPACIDAD FORMATIVA *TRAINING CAPACITY*

Not Applicable

7. CONDICIONES ESPECÍFICAS PARA LA EJECUCIÓN DE DETERMINADOS PROYECTOS *SPECIFIC CONDITIONS FOR THE EXECUTION OF CERTAIN PROJECTS*

Not Applicable